

Adroit Technologies

Protocol Driver

Name	Hitachi H Series Serial
DLL	HITACHI.H



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1. Overview

Adroit supports communication with the following Hitachi H Series PLCs: EH150, H200, H250, H252/B/C, H300, H302, H700, H702, H1002, H2000, H2002 and H4010.

2. Wiring

Connecting to CPU port (15 pin) to PC (25 pin) [DTE <-> DTE]

PLC 15 pin	Designation			PC 25 pin	PC 9 pin
2	Tx	→	RX	3	2
3	RX	←	TX	2	3
4	RS (a)	→	CTS	4	8
5	CS (b)	←	RTS	5	7
6	RV1	→	DSR	6	6
		→	CD	8	1
7	DR (c)	←	DTR	20	4
8	PHL (d)				
9,10	Sig. Ground		Sig. Ground	7	5
11,12	5V Out		NC		
13	-12 Out		NC		
14	+12 Out		NC		

- (a) High when CPU is ready to receive
- (b) Must be high to allow CPU to send (may be fixed high)
- (c) Must be high when the host is ready
- (d) Set high to ignore dipswitch and select 19200 baud speed.

Connecting to CPU port (15 pin) to MODEM (25 pin) [DTE<->DCE ~ DCE <-> DTE]

PLC 15 pin	Designation			Modem 25 pin	PC 9 pin
2	Tx	→	Tx	2	3
3	RX	←	Rx	3	2
4	RS (a)	→	RTS	5	7
5	CS (b)	←	CTS	4	8
6	RV1	→	DTR	20	4
			CD	8	1
7	DR (c)	←	DSR	6	6
8	PHL (d)				
9,10	Sig. Ground		Sig. Ground	7	5
11,12	5V Out		NC		
13	-12 Out		NC		
14	+12 Out		NC		

2.1.1. For Modems

Connect PLC via the [DTE<->DCE] cable to the receiving Modem.

Connect PC via a straight modem cable (usually ships with modem) to the Transmitting modem.

Recommended US Robotics Modem config Setups:

Configuring PC-side modem use Hyper-terminal:

Start with "AT&F0" factory reset. "ATI4" displays current configuration. Alternatively, insert this string into device modem configuration for auto loading:

"ATE0&B1&N10&C1&D2&R2"

Save with "AT&W0", "AT&W1".

&B1, &N10 forces it to connect at only 19200.

Careful: to match your configuration, see your modem documents for specific details.

&D2, &R2 forcing hang-ups, etc., during communications attachment.

Configuring PLC-side modem use Hyper-terminal:

"ATE0S0=1&B1&N10&C1&D0&H1&S1&R2"

"&D0" forces modem to accept incoming connections without terminal ready (TR).

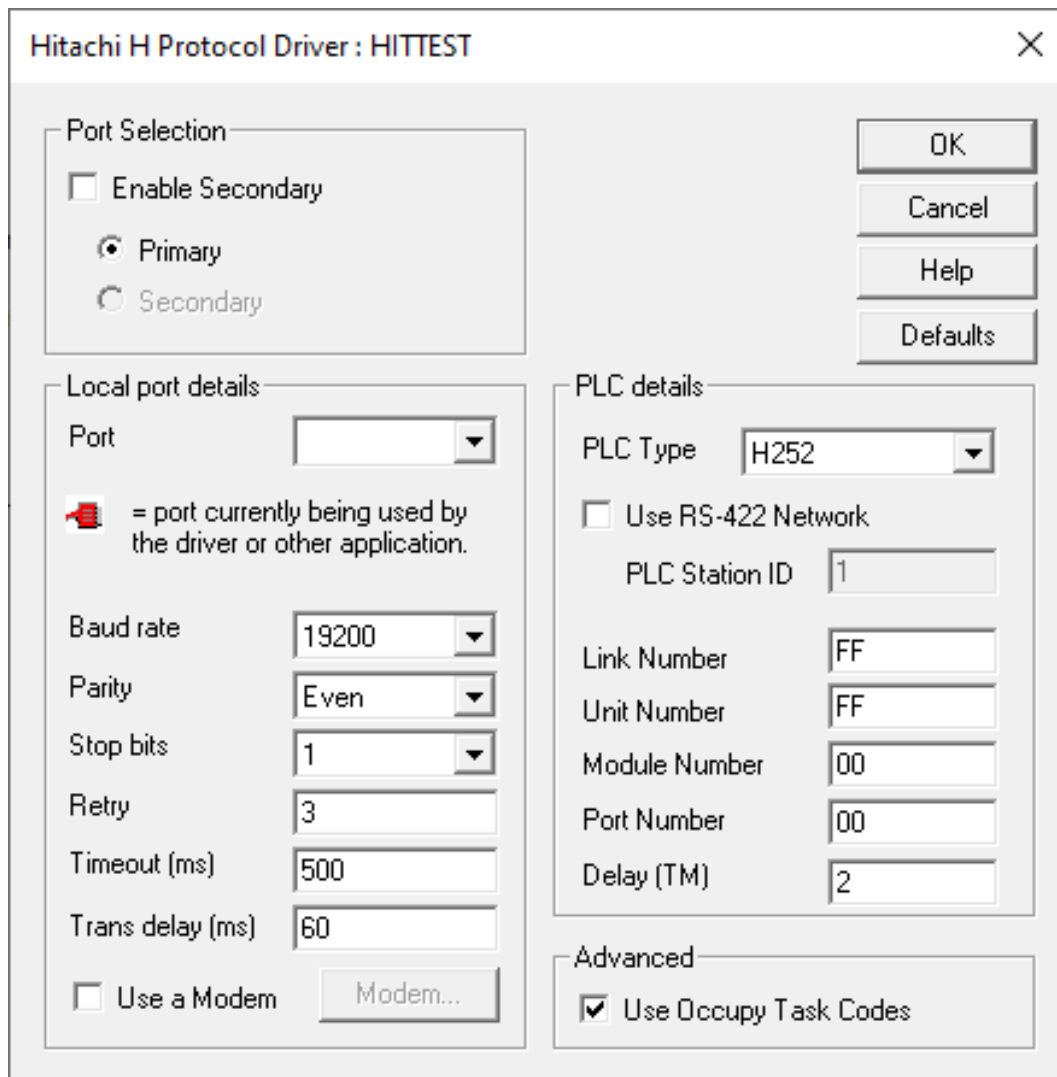
"&H1", "&S1", "&R2" enables handshaking to the PLC.

"S0=1" auto answer mode.

"&B1", "&N10" forces it to connect at only 19200 (**Careful:** make sure to match your device configuration.)

3. Device Configuration

3.1. Device Configuration



The image shows a screenshot of the 'Hitachi H Protocol Driver : HITTEST' configuration window. The window is divided into several sections:

- Port Selection:** Contains a checkbox for 'Enable Secondary'. Below it are two radio buttons: 'Primary' (selected) and 'Secondary'.
- Local port details:** Includes a 'Port' dropdown menu, a red icon with a text explanation '= port currently being used by the driver or other application.', 'Baud rate' (19200), 'Parity' (Even), 'Stop bits' (1), 'Retry' (3), 'Timeout (ms)' (500), 'Trans delay (ms)' (60), and a checkbox for 'Use a Modem' with a 'Modem...' button.
- PLC details:** Includes a 'PLC Type' dropdown (H252), a checkbox for 'Use RS-422 Network', 'PLC Station ID' (1), 'Link Number' (FF), 'Unit Number' (FF), 'Module Number' (00), 'Port Number' (00), and 'Delay (TM)' (2).
- Advanced:** Contains a checked checkbox for 'Use Occupy Task Codes'.

On the right side of the window, there are buttons for 'OK', 'Cancel', 'Help', and 'Defaults'.

3.1.1. Port Selection

Enable Secondary - (Default = unchecked) Enables secondary configuration for dual redundant devices via **Primary** and **Secondary** radio buttons. The secondary device will be connected to if Adroit fails to communicate with the Primary device.

Do not enable if a secondary channel is not configured correctly.

3.1.2. Local port details

Port - The available serial ports on your machine. An icon next to the port name indicates status, " for example shows that the port is currently in use. This may mean your Agent Server is already running, or that another application or device is currently using the port.

Baud rate - The speed of data transmission. Any of the following Baud Rates may be specified: 110, 300, 600, 1200, 2400, 4800, 9600, 19200 or 64800.

Parity - The data parity used in the data transmission. This can be None, Odd or Even.

Stop bits - Forms part of the serial communication setting. For example, 8 data bits, 1 or 2 Stop bits, no parity serial communication. Ensure that the Stop bit settings are the same on the local and remote machines. (Data bits are fixed to 7)

Retry - The number of times that Adroit will try to gain communication with a PLC before failing a transaction.

Timeout (ms) - The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write).

Trans delay (ms) – (Default = 60) If non-zero CTS (Clear-to-send) flow control is enabled, and the driver uses this value as a “settle” time for the outgoing CTS signal whenever it is about to send any requests.

3.1.3. PLC Details

PLC Type - Applicable Hitachi PLC types (as detailed above.)

Use RS-422 Network – Uncheck/disable for point-to-point connection. Check/enable for RS-422 multi-dropped connection.

PLC Station ID – (RS-422 only) station ID as set on PLC.

Link Number – (Default = FF) Link # in hexadecimal.

Unit Number – (Default = FF) Unit # in hexadecimal.

Module Number – (Default = 00) Module version number.

Port Number – (Default = 00) Port # in hexadecimal.

Delay (TM) – (Default = 2) The TM (TW) delay in hexadecimal.

3.1.4. Advanced

Use Occupy Task Codes – (Default = unchecked) This option will be set depending on CPU model selected. If it was automatically set for a CPU, this option will be disabled but “ON”.

3.1.5. Defaults Button

Load Default values – Will reset all settings shown to what they were before this device was created.

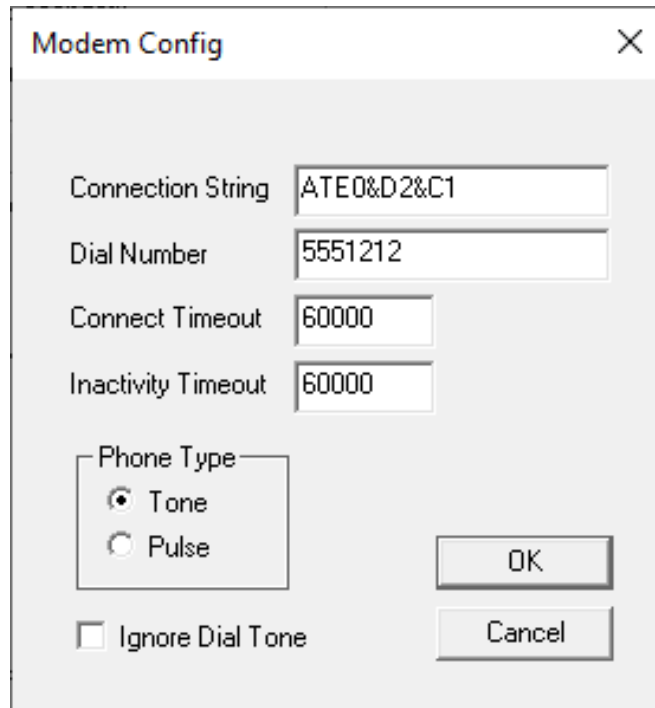
Specific CPUs only operate correctly with special parameter changes viz.

H300, H302, H700 and H702 models require that the Trans delay parameter be adjusted to 60ms or more. (Rev 20c onwards of the driver does this).

Models H200 H300 H307 H700 and H2000 require the Task codes option turned on. In this case, the driver does not allow the user to manipulate the setting either.

3.1.6. Use a Modem & Modem button

Use Modem – (Default = unchecked). Will allow the user to configure a modem for dialup.



The image shows a 'Modem Config' dialog box with the following fields and options:

- Connection String:** ATE0&D2&C1
- Dial Number:** 5551212
- Connect Timeout:** 60000
- Inactivity Timeout:** 60000
- Phone Type:**
 - ☒ Tone
 - ☐ Pulse
- ☐ Ignore Dial Tone
- Buttons:** OK, Cancel

Connection String – Runtime string to upload to the modem on every communication attach. You do not have to specify a save on this string as it is sent every time.

Dial Number – Device Telephone number. For example: (011,555,1212) where ',' is a pause.

Connect Timeout – (Default = 60000ms) Time delay to wait while trying to establish a connection before retrying.

Inactivity Timeout – (Default = 60000ms) Time delay to hang-up after port goes silent.

Phone Type – (Default = Tone) Tone or Pulse.

Ignore Dial Tone – Select if you suspect voltage problems on the telephone line.



PLEASE NOTE

Modem dialup is supported for solicited communication only. If a second device is attached in the driver and its telephone number is identical the modem, it will not be hung up before an attempt at communications is made, which is great for clustered RTU's connected to one modem. However, if the telephone number is different in any way (including pauses, etc.) then the modem will hang up if comms is requested to another device.

4. Supported Addresses

Hitachi PLC data address types supported by the Hitachi Adroit Protocol driver

Device	PLC Data Type	Boolean	Integer	Real
R Area	Digital	R000	-	-
M Area	Digital	M000	-	-
L Area	Digital	L000	-	-
WR Area	Unsigned 16bit Reg		WR0000	WR0000
	Signed 16bit Reg		WR0000I	WR0000I
WM Area	Unsigned 16bit Reg		WM0000	WM0000
	Signed 16bit Reg		WM0000I	WM0000I
WL Area	Unsigned 16bit Reg		WL0000	WL0000
	Signed 16bit Reg		WL0000I	WL0000I
DR Area	Unsigned 32bit Reg		WR0000	DR0000
	Signed 32bit Reg		WR0000L	DR0000L
	Float 32bit Reg		DM0000R	DM0000R
DM Area	Signed 32bit Reg		DM0000	DM0000
	Unsigned 32bit Reg		DL0000L	DL0000L
	Float 32bit Reg		DL0000R	DL0000R
DL Area	Signed 32bit Reg		DM0000	DM0000
	Unsigned 32bit Reg		DL0000L	DL0000L
	Float 32bit Reg		DL0000R	DL0000R



PLEASE NOTE

BOOLEAN types can only be Read from and Written to Digital areas in the PLC (i.e. R, M, L types).

The maximum number of items (digital bits or "16-bit" Registers, with 2 consecutive registers for Doubles) that can be communicated for the different Hitachi PLC's is shown below:

	390	490	590	690
R	64	2000	800	800
M	64	2000	800	800
L	4	125	100	100
WR	32	125	100	100
WM	32	125	100	100
WL	32	125	100	100
DR	16	62	50	50
DM	16	62	50	50
DL	16	62	50	50

5. Protocol Definition and Considerations

5.1. Message Protocol

5.1.1. Analog or Digital Areas

Read Request:

	Lump Address	Function	I/O Code	Start	No Points	Checksum	
<STX>	FFFF0000	40	0A	000000	00	xx	<CR>

PLC Acknowledgement

ACK/NAK	CR
06	<CR>

Adroit ENQ

ENQ	CR
05	<CR>

Read Response:

	Lump Address	Exception Resp	Function	Data	Data	Checksum	
<STX>	FFFF0000	00	02	02	0F	xx	<CR>

Write Request:

	Lump Address	Function	I/O Code	Start Address	Force Data	Checksum	
<STX>	FFFF0000	42	0A	000000	FFFF	xx	<CR>

PLC Acknowledgement

ACK/NAK	CR
06	<CR>

Adroit ENQ

ENQ	CR
05	<CR>

Write Response:

	Lump Address	Exception Resp	Function	Checksum	
<STX>	FFFF0000	00	42	xx	<CR>

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting Notes

1. **H200** type PLC's require CTS handshaking. This is automatically enabled if the PLC Type is H200, but not for other types.
2. A **Transaction Delay** value of 60-100ms has been found to suit most applications. It is required as some Hitachi PLC's cannot handle too many transactions in quick succession. Check the Windows NT event log to see if you are getting too many retries or timeouts and try adjusting the Transaction Delay value to optimise performance. The default setting here is 60 milliseconds, newer controllers may be able to use 20 ms. Adjust this setting until stable comms is achieved for maximum performance.
3. Another factor sometimes causing **excessive errors/retries** in the H200/250/252 PLC types is if the PLC program is too short. Some users have reported much more reliable communications to these PLC's simply by increasing the PLC program size.
4. If you get **intermittent comms failures** on older models ... try setting the tm delay to 2. This delay's default value is 2, a value of 0 may be used for some newer CPU models.

7. Appendix A – Abbreviations

7.1. Abbreviations

ACK	H06	Acknowledgement (message delivered ok)
NAK	H15	Negative acknowledgement (message delivery failed)
ENQ	H05	Enquire
EOT	H04	End of Text
STX	H02	Start of Text
RTC	Return code	
TM	(Response TM) Time for waiting for tasks from the PC	

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	17-Jan-2023	K. Robinson	Grammar and spelling edits, alignment and general editing.	D. Jordaan
2.0				
3.0				

8.2. DLL Revision

Rev.	Date	Changes
	21-Nov-00	Re-formatted margins and styles.
	15-Mar-01	Dialog-box updated and described after it was fixed in rev .
	6-Mar-02	Modem support added.
	06-Apr-02	Protocol description enhanced; Abbreviations added.
	26-Jan-04	Doc styles & style updated.
	5-Feb-04	Setup dialog has changed.
	18-Feb-04	Setup dialog catering for task-codes and the Trans-delay "auto-settings" documented.
	27-Feb-04	Removed model H1000, updated model list, default settings changed for rev 21. Header & footers fixed.
25	09-May-18	Added to Adroit 10.

